

Vydyne R515H NT is a general purpose, 15% glass-filled, heat-stabilized PA66 based resin designed for injection molding applications. R515H NT offers improved flow with a natural surface finish and maintains the excellent resistance typical of PA66 in chemicals, machine and motor oils, solvents, and gasoline.

General

Regional Availability	• North America	• Europe	• Asia Pacific
Additive	• Heat Stabilizer	• Lubricant	
Features	• Chemical Resistant • Grease Resistant • High Strength • Solvent Resistant	• Gasoline Resistant • High Flow • Lubricated	• Good Mold Release • High Rigidity • Oil Resistant
Agency Rating	• ASTM, D4066 PA012G15 • EU, 10/2011 • SAE, J1639 PA1112	• ASTM, D6779 PA012G15 • EU, 2023/2006	• EC, 1935/2004 • FDA, 21 CFR 177.1500
Automotive Specifications	• GM GMW3038P-PA66-GF15H		
UL File Number	• E70062		
Certifications	• Food approval 10/2011	• Food approval FDA 21 CFR	
Appearance	• Natural Color		
Forms	• Pellets		
Processing Method	• Injection Molding		

Physical	dry	cond.	Unit	Test Standard
Density	1.24	-	g/cm ³	ISO 1183
Molding Shrinkage				ISO 294-4
Across Flow : 23°C, 2.00 mm	1.0	*	%	
Flow : 23°C, 2.00 mm	0.5	*	%	
Water Absorption				ISO 62
23°C, 24 hr	1	*	%	
Equilibrium, 23°C, 50% RH	2.2	*	%	

Mechanical	dry	cond.	Unit	Test Standard
Tensile Modulus (23°C)	6600	4000	MPa	ISO 527-2
Tensile Stress (Break, 23°C)	120	80	MPa	ISO 527-2
Tensile Strain (Break, 23°C)	3	13	%	ISO 527-2
Flexural Modulus (23°C)	5900	3300	MPa	ISO 178
Flexural Strength (23°C)	170	110	MPa	ISO 178
Poisson's Ratio (23°C)	0.4		-	ISO 527-2

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Impact	dry	cond.	Unit	Test Standard
Charpy Notched Impact Strength				ISO 179/1eA
+23°C	6	7.5	kJ/m ²	
-30°C	6	5.3	kJ/m ²	
Charpy Unnotched Impact Strength				ISO 179/1eU
+23°C	39	43	kJ/m ²	
-30°C	32	38	kJ/m ²	
Notched Izod Impact Strength				ISO 180/1A
+23°C	6.6	7	kJ/m ²	
-30°C	5.7	5.4	kJ/m ²	

Thermal	dry	cond.	Unit	Test Standard
Heat Deflection Temperature				ISO 75-2/A
1.80 MPa, Unannealed	241	-	°C	
0.45 MPa, Unannealed	258	-	°C	
Melting Temperature	260	*	°C	ISO 11357-3
CLTE				ISO 11359-2
Flow : 23 to 55°C, 2.00 mm	30	*	E-6/K	
Transverse : 23 to 55°C, 2.00 mm	112	*	E-6/K	
RTI Elec				UL 746
0.750 mm	140		°C	
1.50 mm	140		°C	
3.00 mm	140		°C	
RTI Imp				UL 746
0.750 mm	120		°C	
1.50mm	120		°C	
3.00 mm	120		°C	
RTI Str				UL 746
0.750 mm	125		°C	
1.50 mm	140		°C	
3.00 mm	140		°C	

Electrical	dry	cond.	Unit	Test Standard
Volume Resistivity (1.00 mm)	1E11	-	Ohm*m	IEC 60093
Dielectric Strength (1.00 mm)	20	-	kV/mm	IEC 60243
Arc Resistance (3.00 mm)	6		-	ASTM D 495
High Amp Arc Ignition (HAI)				UL 746
0.750 mm	PLC 0		-	
1.50 mm	PLC 0		-	

Vydyne® R515H NT

polyamide 66



3.00 mm	PLC 0	-	
High Voltage Arc Tracking Rate (HVTR), 3.00 mm	PLC 1	-	UL 746
Hot-wire Ignition (HWI)			UL 746
0.400 mm	PLC 4	-	
0.750 mm	PLC 3	-	
1.50 mm	PLC 4	-	

Flammability	Value	Unit	Test Standard
Flammability			UL 94
0.750 mm	HB	-	
1.50 mm	HB	-	
3.00 mm	HB	-	
Glow Wire Flammability Index			IEC 60695-2-12
0.400 mm	675	°C	
0.750 mm	675	°C	
1.50 mm	675	°C	
Glow Wire Ignition Temperature			IEC 60695-2-13
0.400 mm	700	°C	
0.750 mm	700	°C	
1.50 mm	700	°C	

Injection	Value	Unit
Drying Temperature	80	°C
Drying Time	4	h
Rear Temperature	280 - 310	°C
Middle Temperature	280 - 310	°C
Front Temperature	280 - 310	°C
Nozzle temperature	280 - 310	°C
Processing (Melt) Temperature	285 - 305	°C
Mold Temperature	65 - 95	°C



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